

Math 196 Problem Set 8

Due May 21, at 10:00 PM

Problem 1. Interpret the area of the parallelogram with the following points as its vertices as the determinant of a matrix, and use this to compute it.

$$(0, 0), (2, -1), (1, 3), (-1, 4)$$

Problem 2. List all permutations of length 3, and determine the sign of each. Using these, write down the formula for

$$\det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

and check that you get the same terms by extending the matrix and drawing diagonal lines as we did in class.

Problem 3. Use determinants to decide if each of the following matrices is invertible.

$$\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 3 & 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{bmatrix}$$

Problem 4. Compute the determinants of the following matrices using cofactor expansion.

$$\begin{bmatrix} 1 & -1 & 2 & 0 \\ 1 & 2 & -4 & 1 \\ 2 & 0 & 0 & -1 \\ 0 & 1 & 2 & -3 \end{bmatrix}, \begin{bmatrix} 1 & 2 & -3 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 2 & 3 & 4 \\ 0 & 0 & 2 & 2 \end{bmatrix}$$

Problem 5. Compute the determinants of the following matrices using row operations.

$$\begin{bmatrix} -2 & 2 & 4 & -2 \\ 1 & -1 & 2 & 0 \\ 1 & 0 & 3 & 1 \\ 2 & 3 & 1 & 4 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & -3 \\ 0 & 0 & 2 & 1 \\ 0 & 1 & 3 & 1 \\ 4 & 2 & 5 & 3 \end{bmatrix}$$

Problem 6. This problem goes through an argument that for $n \times n$ matrices A and B , $\det(AB) = \det(A)\det(B)$.

- (a) Check that $\det(AB) = \det(A)\det(B)$ in the case that B is not invertible.
- (b) Suppose that f is a function assigning a number $f(A)$ to each $n \times n$ matrix A , and that f satisfies the following properties:
 - (i) If B is obtained by swapping two rows of A , then $f(B) = -f(A)$.
 - (ii) If B is obtained by multiplying a row of A by k , then $f(B) = kf(A)$.
 - (iii) If B is obtained by adding a multiple of one row of A to another row, then $f(B) = f(A)$.
 - (iv) $f(I_n) = 1$, and $f(A) = 0$ if A is not invertible.

Argue that $f(A) = \det(A)$ for all matrices A .¹

¹Think about the process of bringing A to reduced row echelon form.

(c) Suppose that B is invertible. Check that the function

$$f(A) = \frac{\det(AB)}{\det(B)}$$

has properties (i) - (iv).² Conclude from this that $\det(AB) = \det(A)\det(B)$ in this case as well.

Problem 7. Each of the following matrices has a single eigenvalue, 2. For each one, find a basis of the eigenspace V_2 .

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

For the next problem, you can use the following fact, which I will prove on Tuesday.

Proposition. Suppose that $\lambda_1, \dots, \lambda_r$ are distinct eigenvalues of a matrix A , and for each i , $v_i \in V_{\lambda_i}$. If

$$v_1 + \dots + v_r = 0$$

then $v_i = 0$ for all i .

Problem 8. Suppose that $\lambda_1, \dots, \lambda_r$ are distinct eigenvalues of a matrix A , and that for each i , $v_{i,1}, \dots, v_{i,d_i}$ is a basis for V_{λ_i} . Prove that all of these vectors together:

$$v_{1,1}, \dots, v_{1,d_1}, v_{2,1}, \dots, v_{2,d_2}, \dots, v_{r,1}, \dots, v_{r,d_r}$$

form a linearly independent set.

Problem 9 (Optional). Suppose that A is a symmetric matrix, meaning $A^\top = A$. Let v and w be eigenvectors for distinct eigenvalues $\lambda \neq \mu$, so $Av = \lambda v$, and $Aw = \mu w$. Show that v and w are orthogonal.³

Problem 10. For each matrix A , find its (real) eigenvalues. For each eigenvalue, find a basis of the corresponding eigenspace. If these bases together form a basis for $\mathbb{R}^n$ (see Problem 8), use this to write down a diagonalization of A , of the form $A = SDS^{-1}$ where D is diagonal and S invertible.

(a) $\begin{bmatrix} 6 & 3 \\ 2 & 7 \end{bmatrix}$

(b) $\begin{bmatrix} 4 & 5 \\ -2 & -2 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}$

(d) $\begin{bmatrix} -1 & 2 & 1 \\ -1 & 2 & 1 \\ -2 & 2 & 2 \end{bmatrix}$

(e) $\begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{bmatrix}$

Problem 11. Suppose that a matrix P represents orthogonal projection onto a subspace $W \subset \mathbb{R}^n$. What are the eigenvalues of P ? What are the corresponding eigenspaces?

²Hint: If we apply a row operation to A , what happens to AB ?

³Hint: consider $v^\top Aw$.